

PART-1 Pen and Paper

o Problem: 1 :

Given;

$a, b, c \in \mathbb{R}^3$ and a, b, c are 3 vectors

Verify

a) $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$ (No AI used)

let suppose, as given a, b, c are arbitrary vector

$\therefore a = \{a_x \hat{i} + a_y \hat{j} + a_z \hat{k}\}$

$b = \{b_x \hat{i} + b_y \hat{j} + b_z \hat{k}\}$

$c = \{c_x \hat{i} + c_y \hat{j} + c_z \hat{k}\}$

LHS:

So as given;

Taking

$(\vec{b} \times \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} \left[\begin{array}{l} \text{can be written in} \\ \text{matrix} \\ \text{form} \end{array} \right]$

$\Rightarrow$ Solving Determinant we get

$(\vec{b} \times \vec{c}) = \hat{i} b_x [c_y \hat{j} + b_z \hat{k} - b_y \hat{j} \cdot c_z \hat{k}]$
 $- \hat{j} b_y [b_x \hat{i} c_z \hat{k} - b_z \hat{k} c_x \hat{i}]$
 $+ \hat{k} b_z [b_x \hat{i} c_y \hat{j} - b_y \hat{j} c_x \hat{i}]$

So as we can exclude the unit vectors

$(\vec{b} \times \vec{c}) = \underbrace{\vec{b}_x [\vec{c}_y \vec{b}_z - \vec{b}_y \vec{c}_z]}_{x \text{ coordinate}} - \underbrace{\vec{b}_y [\vec{b}_x \vec{c}_z - \vec{b}_z \vec{c}_x]}_{y \text{ coordinate}} + \underbrace{\vec{b}_z [\vec{b}_x \vec{c}_y - \vec{b}_y \vec{c}_x]}_{z \text{ coordinate}}$

as $\vec{a} \cdot (\vec{b} \times \vec{c}) \Rightarrow a = \{a_x, a_y, a_z\}$

Multiplying the same coordinate as from a to $(\vec{b} \times \vec{c})$ being a dot product.

$$\therefore \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$\Rightarrow a_x(b_y c_z - b_z c_y) - a_y(b_x c_z - b_z c_x) + a_z(b_x c_y - b_y c_x)$$

$$= a_x b_y c_z - a_x b_z c_y - a_y b_x c_z + a_y b_z c_x + a_z b_x c_y - a_z b_y c_x$$

As we are targeting a vector x to dot product c at RHS.

$\therefore$ Taking same coordinates of c common

So making C_x, C_y, C_z coordinates

$$\rightarrow (a_y b_z - a_z b_y) C_x - C_y (a_x b_z - a_z b_x) + C_z (a_x b_y - a_y b_x)$$

$$= \begin{vmatrix} \overset{\downarrow}{i} & \overset{\downarrow}{j} & \overset{\downarrow}{k} \\ a_y & b_y & c_y \\ a_z & b_z & c_z \end{vmatrix} C_x - \begin{vmatrix} \overset{\downarrow}{i} & \overset{\downarrow}{j} & \overset{\downarrow}{k} \\ a_x & b_z & c_z \\ a_y & b_x & c_x \end{vmatrix} C_y + \begin{vmatrix} \overset{\downarrow}{i} & \overset{\downarrow}{j} & \overset{\downarrow}{k} \\ a_x & b_y & c_y \\ a_y & b_x & c_x \end{vmatrix} C_z$$

$$= \{C_x, C_y, C_z\} \cdot \begin{bmatrix} \overset{i}{a_x} & \overset{j}{a_y} & \overset{k}{a_z} \\ \overset{i}{b_x} & \overset{j}{b_y} & \overset{k}{b_z} \end{bmatrix}$$

$$= \vec{c} \cdot [\vec{a} \times \vec{b}]$$

$$= \text{RHS} \rightarrow [\vec{a} \times \vec{b}] \cdot \vec{c}$$

Hence proved.

2. $\boxed{a \times (b \times c) = (a \cdot c)b - (a \cdot b)c}$ (No A used)

(b)

As solved in the previous case

I can directly write now cross product of

$$(\vec{b} \times \vec{c}) = \hat{i}(b_y c_z - b_z c_y) - \hat{j}(b_x c_z - b_z c_x) + \hat{k}(b_x c_y - b_y c_x)$$

①

where;

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{c} = c_x \hat{i} + c_y \hat{j} + c_z \hat{k}$$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

Considering $(\vec{b} \times \vec{c}) = \vec{x}$ where $x \Rightarrow 1$

$\therefore$ on LHS side.

we can write

$$\rightarrow \vec{a} \times (\vec{b} \times \vec{c}) = \vec{a} \times \vec{x}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_y c_z - b_z c_y & -(b_x c_z - b_z c_x) & b_x c_y - b_y c_x \end{vmatrix}$$

Taking determinant for only $\boxed{x \text{ components}}$

$$\Rightarrow a_y [b_x c_y - b_y c_x] + a_z (b_x c_z - b_z c_x)$$

$$\Rightarrow b_x (a_y c_y + a_z c_z) - c_x (a_y b_y + a_z b_z) \quad (\because \text{Taking } b_x \text{ and } c_x \text{ common.})$$

adding $a_x c_x b_x$ term as RHS $(a \cdot c)b$ ~~so~~ ^{component} so x has $a_x b_x c_x$ term as we are adding we can subtract as well.

$$\Rightarrow b_x (a_y c_y + a_z c_z) - c_x (a_y b_y + a_z b_z) + (a_x b_x c_x - a_x b_x c_x)$$

$$\Rightarrow b_x (a_y c_y + a_x c_x + a_z c_z) - c_x (a_y b_y + a_z b_z + a_x b_x)$$

$$\Rightarrow b_x (\vec{a} \cdot \vec{c}) - c_x (\vec{a} \cdot \vec{b})$$

Similarly for y & z components

$$b_y (\vec{a} \cdot \vec{c}) - c_y (\vec{a} \cdot \vec{b})$$

$$b_z (\vec{a} \cdot \vec{c}) - c_z (\vec{a} \cdot \vec{b})$$

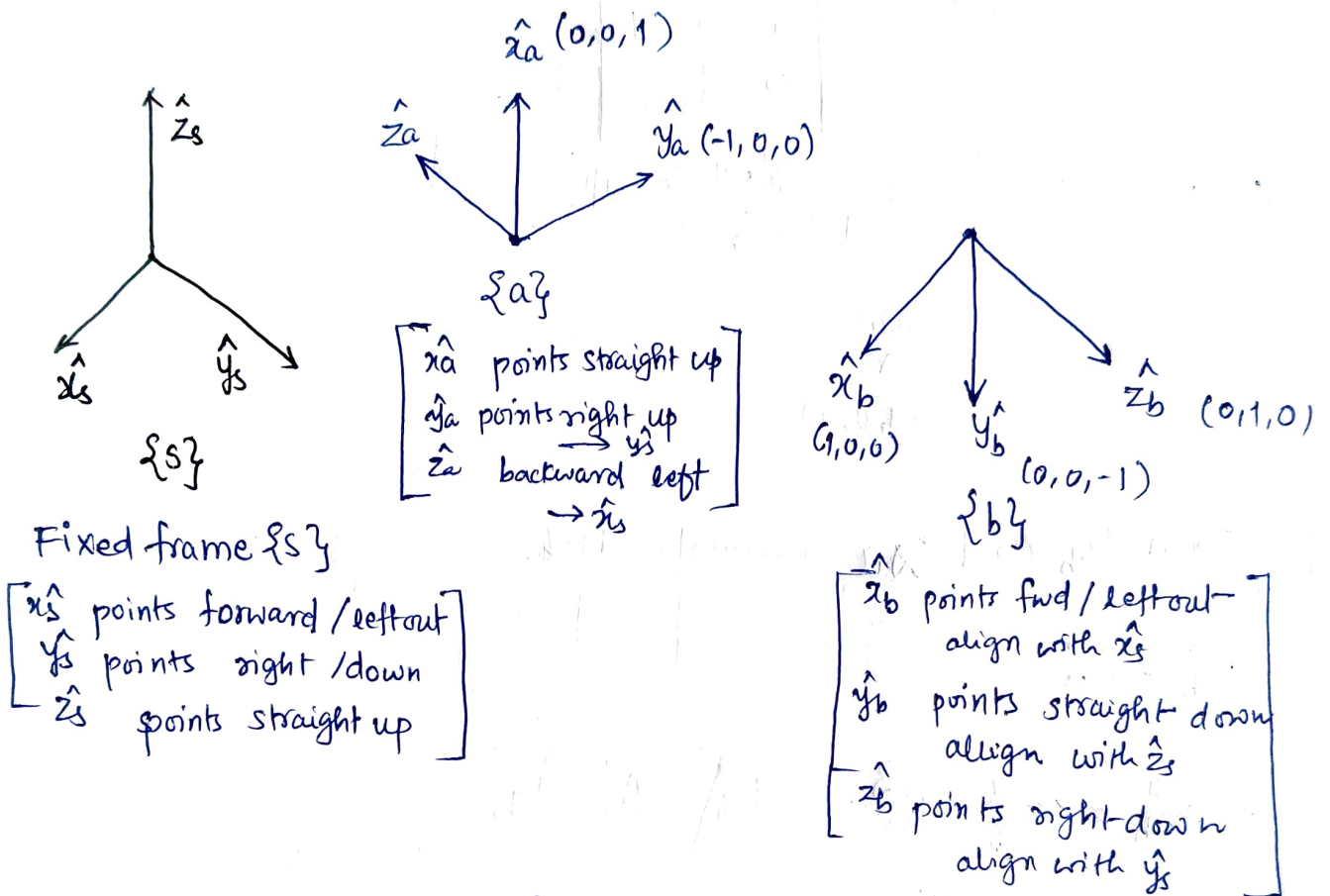
Combining 3 components

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{b}_x + \vec{b}_y + \vec{b}_z) (\vec{a} \cdot \vec{c}) - (\vec{c}_x + \vec{c}_y + \vec{c}_z) (\vec{a} \cdot \vec{b})$$
$$= \vec{b} (\vec{a} \cdot \vec{c}) - \vec{c} (\vec{a} \cdot \vec{b})$$

$\Rightarrow$ RHS. Hence proved.

Problem: 2

(a) Draw free hand three frame at different location



(b) Rotation Matrix for each frame.

$$R_{a^s} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$R_a^s = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_b^s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

where

$$x_a^s = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad x_b^s = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

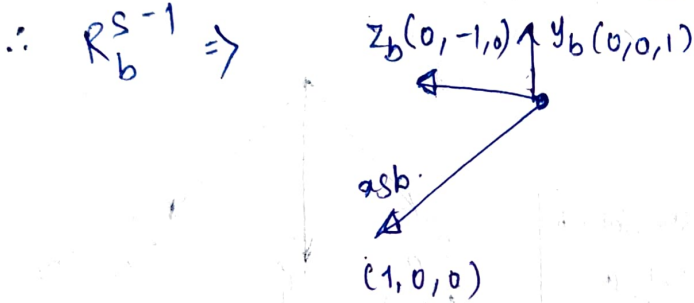
$$y_a^s = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad y_b^s = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$z_a^s = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad z_b^s = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\leftrightarrow$
opp direction

(c) Given R_b^S calculate $R_b^{S^{-1}}$ without matrix inverse

$$(R_b^S)^{-1} = (R_b^S)^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$



(d) Given R_b^S and R_a^S calculate R_b^a .

$$R_b^S \cdot R_a^S$$

$$\therefore \cancel{R_b^a} = \cancel{R_a^S} \cdot \cancel{R_b^S} \quad R_b^a = R_a^S \cdot R_b^S$$

$$\therefore R_b^a = (R_a^S)^{-1} R_b^S$$

$$R_b^a = (R_a^S)^T R_b^S$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_b^a = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$(e) R_1 = R_a^S \cdot R$$

$$= R_a^S \cdot R_b^S$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_1 \leftrightarrow R_a^S (-90^\circ)$ about body fixed $\hat{x}_a$ axis

$$R_2 = R \cdot R_a^S$$

$$= R_b^S \cdot R_a^S$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 \leftrightarrow R_a^S (-90^\circ)$ about ground / fixed frame $\hat{x}_g$ axis

$$(f) P_b = (1, 2, 3)$$

$$P_s = R_b^S \cdot P_b^T$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$P_s = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

$$(g)$$

$$P'' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} \Rightarrow \text{local transformation}$$

$$P'' = R_{sb}^T P_s$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \Rightarrow \text{coordinate change.}$$

$$(h) \omega_s = (3, 2, 1)$$

$$\omega_a = R_s^a \times \omega_s$$

$$= (R_a^s)^{-1} \times \omega_s$$

$$= [R_a^s]^T \times \omega_s$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}^T \times \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\omega_a = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}} = \omega_a$$

Problem: 3 (NO M used)

19,
5.

Fixed frame

$$P_g = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{2}} \right) \quad \{\text{initial point}\}$$

Rotation about x axis $\theta_x = 30^\circ$

$$\downarrow R_x \cdot P$$

Rotation about y axis $\theta_y = 135^\circ$

$$\downarrow R_y (R_x \cdot P)$$

Rotation about z axis $\theta_z = -120^\circ$

$$\downarrow R_z [R_y (R_x \cdot P)] = R$$

So new point will be

$$P' = R P$$

Where $R \Rightarrow$ Rotation Matrix $\Rightarrow$ manipulated as $R_z [R_y (R_x \cdot P)]$ order.

Let's solve inner bracket:

$R_x \cdot P \Rightarrow$ as angle is 30° abt x axis

$$\therefore R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \quad \uparrow$$

at $\theta = 30^\circ$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & -1/2 \\ 0 & 1/2 & \sqrt{3}/2 \end{bmatrix}$$

$$\therefore R_x \cdot P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & -1/2 \\ 0 & 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Now multiplying R_x Matrix by R_y before. at $\theta_y = 135^\circ$

$$R_y \cdot R_x \Rightarrow \begin{bmatrix} \cos(135^\circ) & 0 & \sin(135^\circ) \\ 0 & 1 & 0 \\ -\sin(135^\circ) & 0 & \cos(135^\circ) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & -1/2 \\ 0 & 1/2 & \sqrt{3}/2 \end{bmatrix}$$

$\theta = 135$

$$R_y \cdot R_x = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{4} & \frac{\sqrt{6}}{4} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{6}}{4} \end{bmatrix}$$

Now final rotation abt z axis (-120°)

$$R_z(-120^\circ) \cdot R_y \cdot R_x$$

$$= \begin{bmatrix} \cos(-120^\circ) & -\sin(-120^\circ) & 0 \\ \sin(-120^\circ) & \cos(-120^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/4 & \sqrt{6}/4 \\ 0 & \sqrt{3}/2 & -1/2 \\ -\sqrt{2}/2 & -\sqrt{2}/4 & -\sqrt{6}/4 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/4 & \sqrt{6}/4 \\ 0 & \sqrt{3}/2 & -1/2 \\ -\sqrt{2}/2 & \sqrt{2}/4 & -\sqrt{6}/4 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2}/4 & \frac{3}{4} - \frac{\sqrt{2}}{8} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{6}}{8} \\ \sqrt{6}/4 & -\frac{\sqrt{3}}{4} - \frac{\sqrt{6}}{8} & \frac{1}{4} - \frac{3\sqrt{2}}{8} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{6}}{4} \end{bmatrix} \Rightarrow R$$

$$\therefore P^1 = R P^T$$

$$= \begin{bmatrix} \sqrt{2}/4 & 3/4 - \sqrt{2}/8 & -\sqrt{3}/4 - \sqrt{6}/8 \\ \sqrt{6}/4 & -\sqrt{3}/4 - \sqrt{6}/8 & 1/4 - 3\sqrt{2}/8 \\ -\sqrt{2}/2 & -\sqrt{2}/4 & -\sqrt{6}/4 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{6} \\ 1/\sqrt{2} \end{bmatrix}$$

Ans: B

$$R = R_z(-120^\circ) \cdot R_y(135^\circ) \cdot R_x(30^\circ)$$

$$\begin{bmatrix} 0.3536 & 0.5732 & -0.7392 \\ 0.6124 & -0.7392 & -0.2883 \\ -0.7071 & -0.3536 & -0.6124 \end{bmatrix}$$

$$\therefore P^T = \begin{bmatrix} \sqrt{6}/12 - 3/4\sqrt{6} + \sqrt{2}/16 - \sqrt{6}/8 & -\sqrt{2}/16 \\ \sqrt{2}/4 + \sqrt{18}/24 + \sqrt{6}/48 + \sqrt{2}/8 - 3/8 \\ -\sqrt{6}/6 + \sqrt{12}/24 - \sqrt{12}/8 \end{bmatrix}$$

$$P^T = \begin{bmatrix} -\sqrt{6}/6 & -\sqrt{3}/12 \\ -1/4 + \sqrt{2}/2 \\ -\sqrt{6}/6 - \sqrt{3}/6 \end{bmatrix} \approx \begin{bmatrix} -0.5526 \\ 0.4571 \\ -0.6969 \end{bmatrix} \leftarrow \text{Answer: a}$$

Problem: 4 (No AI used)

9

8.7

As given, there is rotation from frame b to frame a
 $\therefore R_a^b$ will represent the present frame projected on, to the base frame

Similarly

R_b^c rotation from frame c to frame b.

$$\therefore R_a^b = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$R_b^c = \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix}$$

let's evaluate the product of 2 matrix

$$R = R_a^b \cdot R_b^c$$

$$= \begin{bmatrix} r_{11}s_{11} + r_{12}s_{21} + r_{13}s_{31} & r_{11}s_{12} + r_{12}s_{22} + r_{13}s_{32} & r_{11}s_{13} + r_{12}s_{23} + r_{13}s_{33} \\ r_{21}s_{11} + r_{22}s_{21} + r_{23}s_{31} & r_{21}s_{12} + r_{22}s_{22} + r_{23}s_{32} & r_{21}s_{13} + r_{22}s_{23} + r_{23}s_{33} \\ r_{31}s_{11} + r_{32}s_{21} + r_{33}s_{31} & r_{31}s_{12} + r_{32}s_{22} + r_{33}s_{32} & r_{31}s_{13} + r_{32}s_{23} + r_{33}s_{33} \end{bmatrix}$$

Attempt 2 $\rightarrow$ Row 1 x column 1 tracking

$$R_{x1} = \begin{bmatrix} r_{11}s_{11} + r_{12}s_{21} + r_{13}s_{31} \\ r_{21}s_{11} + r_{22}s_{21} + r_{23}s_{31} \\ r_{31}s_{11} + r_{32}s_{21} + r_{33}s_{31} \end{bmatrix}$$

$$R_{x1} = s_{11} \begin{bmatrix} r_{11} \\ r_{21} \\ r_{31} \end{bmatrix} + s_{21} \begin{bmatrix} r_{12} \\ r_{22} \\ r_{32} \end{bmatrix} + s_{31} \begin{bmatrix} r_{13} \\ r_{23} \\ r_{33} \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$
 x_a^b y_a^b z_a^b

$$\therefore \boxed{s_{11} x_a^b + s_{21} y_a^b + s_{31} z_a^b = R_{x1}} = x_a^a \quad x_a^c$$

Similarly for R_{x2} (column: 2)

Side note: Stuckpoint

got stuck solving

rows x column's

index's not matching
now, shifted

to row x row tracking
which maps directly
to physical coordinate
& transformations.

$$R_{*2} = S_{12} \begin{bmatrix} r_{11} \\ r_{21} \\ r_{31} \end{bmatrix} + S_{22} \begin{bmatrix} r_{12} \\ r_{22} \\ r_{32} \end{bmatrix} + S_{32} \begin{bmatrix} r_{13} \\ r_{23} \\ r_{33} \end{bmatrix}$$

$$\therefore R_{*2} = S_{12} x_a^b + S_{22} y_a^b + S_{32} z_a^b = y_a^c$$

$$R_{*3} = S_{13} \begin{bmatrix} r_{11} \\ r_{21} \\ r_{31} \end{bmatrix} + S_{23} \begin{bmatrix} r_{12} \\ r_{22} \\ r_{32} \end{bmatrix} + S_{33} \begin{bmatrix} r_{13} \\ r_{23} \\ r_{33} \end{bmatrix}$$

$$R_{*3} = S_{13} x_a^b + S_{23} y_a^b + S_{33} z_a^b = z_a^c$$

$$\therefore R = \begin{bmatrix} \vdots & \vdots & \vdots \\ R_{*1} & R_{*2} & R_{*3} \\ \vdots & \vdots & \vdots \end{bmatrix}$$

$$= \begin{bmatrix} \vdots & \vdots & \vdots \\ x_a^c & y_a^c & z_a^c \\ \vdots & \vdots & \vdots \end{bmatrix}$$

$$\text{as } \Rightarrow R = R_a^b \cdot R_b^c$$

$$R = R_a^c$$

Problem: 5

9

From the lecture No AI used

(a) Prove:

$$R(\vec{v} \times \vec{w}) = (R\vec{v}) \times (R\vec{w})$$

LHS

We multiplied with the vector $\vec{x}$ to the entire left side. also divide it by x

$$R(\vec{v} \times \vec{w}) \cdot \vec{x} = R(\vec{v}) \times R(\vec{w}) \quad \rightarrow \quad R(\vec{v} \times \vec{w}) \cdot \vec{x} = R(\vec{v}) \times R(\vec{w}) \cdot \vec{x}$$

Solving this LHS. $\boxed{R(\vec{v} \times \vec{w}) \cdot \vec{x}}$

We know as R is orthogonal matrix

$R^T = R^{-1}$ which means we can shift R to other side of dot product.

$$\therefore R(\vec{v} \times \vec{w}) \cdot \vec{x} = (\vec{v} \times \vec{w}) \cdot (R^T \vec{x})$$

now as we know

$$\vec{a} \times \vec{b} \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c}) = \det [a \ b \ c] \quad \text{--- (1)}$$

Similarly here,

$$\therefore (\vec{v} \times \vec{w}) \cdot (R^T \vec{x}) = \det [\vec{v} \ \vec{w} \ R^T \vec{x}] \quad \text{from (1)}$$

$$\therefore \det(R) \cdot \det [\vec{v} \ \vec{w} \ R^T \vec{x}]$$

$$\therefore \det [R\vec{v} \ R\vec{w} \ R(R^T \vec{x})]$$

$$\therefore \det [R\vec{v} \ R\vec{w} \ \vec{x}]$$

$$\therefore \det [R\vec{v} \ R\vec{w} \ \vec{x}] = [(R\vec{v}) \times (R\vec{w})] \cdot \vec{x}$$

$$\therefore \boxed{[(R\vec{v}) \times (R\vec{w})] \cdot \vec{x}}$$

= RHS

hence

$$R(\vec{v} \times \vec{w}) \cdot \vec{x} = (R\vec{v}) \times (R\vec{w}) \cdot \vec{x}$$

$$\checkmark \quad R(\vec{v} \times \vec{w}) = R\vec{v} \times R\vec{w}$$

considering $\left. \begin{array}{l} \det(R) = 1 \\ \therefore 1 \cdot \det[\vec{v} \ \vec{w} \ R^T \vec{x}] \\ \downarrow \\ \det(R) \end{array} \right\}$

(b) To prove;

10.

$$R \hat{\omega} R^T = (\hat{R\omega})$$

let's take an arbitrary vector $\omega = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

so its $\hat{\omega} = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$

or R^3 along x axis

$$R = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R^T = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

LHS

$$\therefore R \hat{\omega} R^T$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -z\sin\theta & -z\cos\theta & x\sin\theta + y\cos\theta \\ z\cos\theta & -z\sin\theta & -x\cos\theta + y\sin\theta \\ -y & x & 0 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -z & y\cos\theta + x\sin\theta \\ z & 0 & y\sin\theta - x\cos\theta \\ -y\cos\theta - x\sin\theta & -y\sin\theta + x\cos\theta & 0 \end{bmatrix}$$

①

Computing RHS

$$R\omega = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \\ z \end{bmatrix} \rightarrow \begin{matrix} x' \\ y' \\ z' \end{matrix}$$

$$\hat{R}\omega = \begin{bmatrix} 0 & -z' & y' \\ z' & 0 & -x' \\ -y' & x' & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -z & x\sin\theta + y\cos\theta \\ z & 0 & -x\cos\theta + y\sin\theta \\ \begin{pmatrix} -x\sin\theta \\ -y\cos\theta \end{pmatrix} & \begin{pmatrix} x\cos\theta \\ -y\sin\theta \end{pmatrix} & 0 \end{bmatrix} \quad \text{--- (2)}$$

Comparing final matrix of LHS = final matrix of RHS
 (1) (2)

Here we can say being matrix similar

$$\boxed{R \hat{\omega} R^T = (\hat{R}\omega)}$$

(c) $R = [r_1 \ r_2 \ r_3]$ $\xrightarrow{\text{LHS}} \{ \text{NO AI} \}$

let;

$$r_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}; \quad r_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}; \quad r_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$$

$$\therefore R = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

$$x_1(y_2 z_3 - z_2 y_3) - y_1(x_2 z_3 - z_2 x_3) + z_1(x_2 y_3 - y_2 x_3) = \det R \quad \text{--- (a)}$$

RHS

$$r_2 \times r_3 = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix} = \begin{bmatrix} y_2 z_3 - z_2 y_3 \\ z_2 x_3 - x_2 z_3 \\ x_2 y_3 - y_2 x_3 \end{bmatrix} \quad \text{--- (1)}$$

$$r_1^T = [x_1 \ y_1 \ z_1] \quad \text{--- (2)}$$

$\therefore r_1^T (r_2 \times r_3)$ Multiply 2 to 1

$$= [x_1 \ y_1 \ z_1] \begin{bmatrix} y_2 z_3 - z_2 y_3 \\ z_2 x_3 - x_2 z_3 \\ x_2 y_3 - y_2 x_3 \end{bmatrix} \Rightarrow \det r_1^T (r_2 \times r_3)$$

$$= x_1(y_2 z_3 - z_2 y_3) - y_1(x_2 z_3 - z_2 x_3) + z_1(x_2 y_3 - y_2 x_3) \quad \text{--- (b)}$$

as $\det R = \det r_1^T (r_2 \times r_3)$ from a) & b)

hence $[r_1 \ r_2 \ r_3] = r_1^T (r_2 \times r_3)$
proved.

(d) for vector space proof

objective:

~~let~~ SO_3 be a subset of vector space proof

let say

vector space be $V_3 \in \mathbb{R}^3$

$\therefore$ (i) Considering a zero matrix. O in $SO(3)$

$$\therefore O^T = O = -O \Rightarrow O \in SO(3) \in \mathbb{R}^3 \in V_3$$

(ii) Considering $\forall A$ vectors & $\forall B$ vectors in SO_3

$$\left. \begin{array}{l} A^T = -A \\ B^T = -B \end{array} \right\} \text{ skewsymmetric confirms in } SO_3 \in V_3$$

$$\therefore (A+B)^T = A^T + B^T = -A - B = -(A+B)$$

$$\text{So } (A+B)^T \text{ belong to } SO_3 \in V_3 \in \mathbb{R}^3$$

(iii) Scalar Multiple also in $SO(3)$

$\forall A$ there is CA to which there is CA^T to which there is $-CA$

C is scalar multiply

$$\therefore -CA \in SO_3 \in V_3 \in \mathbb{R}^3$$

Hence, zero matrix, arbitrary vectors, scalar + additive multipliers all belongs to vector space V_3

For dimension
as we know

$$A \in SO_3 \quad A^T = -A$$

① for all values of A as a_{ij} is equal to $-a_{ji} \quad \forall j \in \{1, 2, 3\} \Rightarrow a_{ji} = -a_{ij}$

② for all $i = j$

$$\text{③ } a_{jj} = -a_{jj} \Rightarrow 2a_{jj} = 2a_{ii} = 0$$

$i \neq j$

let's say

$$a_{33} = x; \quad a_{23} = y; \quad a_{21} = z \Leftrightarrow a_{23} = -x; \quad a_{12} = -y; \quad a_{31} = -z$$

$$\therefore A = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$$

$$\therefore \text{Span of } (A) = SO(3)$$

$$\text{dimension } (SO_3) = \dim(A) = 3 \times 3 = 3$$

Let a general matrix $R \in SO(2)$ be defined in a matrix as

$$R = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

of $SO(2)$ is there

2 properties always remains align:

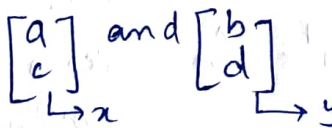
- $R^T R = I$ {orthogonality}
- $\det(R) = \pm 1$ {determinant as unity}

Let's find

$$R^T R = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} \Rightarrow I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

∴ $a^2 + c^2 = 1$ - ① for this set
 $b^2 + d^2 = 1$ - ② of equations.
 $ab + cd = 0$ - ③ ~~eqn to find~~
~~two unit circle~~
~~with $a = \cos \theta$ & $b = \sin \theta$~~

from eq: 3
 $ab + cd = 0$.

of the vector $\begin{bmatrix} a \\ c \end{bmatrix}$ and $\begin{bmatrix} b \\ d \end{bmatrix}$

 are orthogonal & both have det 1.

∴ $x \times y = 90^\circ \rightarrow$ orthogonal plane.

∴ $a = d$
 of $a = d$ from eq 3 $\Rightarrow b = -c$ — ④
 or

$b = c$ " $\Rightarrow d = -a$ — ⑤

so of 4. is there $a = d; b = -c$ CASE: 1

$\det(R) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = a^2 + b^2 + d^2 = 1 \rightarrow$ form 2

$(-c)(-c) + a^2 = 1$

∴ $a^2 + c^2 = 1$ $\xrightarrow{\text{proves}}$ eq: 1 as well

Side note: {stuck point}

I could parametrized
 as trigno functions
 like $a = \cos \theta, b = \sin \theta$
 etc. I wasn't sure
 how to enforce
 $\det(R) = 1$

∴ Instead guessing
 angle
 from eq: 3
 $ab + cd$

of eqⁿ 5 is there.

$$b=c \text{ \& } d=-a.$$

$$\det(R) = a(-a) - b(b) = -(a^2+b^2) = -1.$$

$\therefore$ Case 2 gives -1 & Case 1 gives 1

$$R = \begin{bmatrix} a & -c \\ c & a \end{bmatrix} \xrightarrow{\text{determinant}} (a)$$

The constraint eqⁿ $a^2+c^2=1$ is definition of unit circle $s^1 \in \mathbb{R}^2$ on the point (a,c)

$\therefore$ we know $a=\cos\theta$; $c=\sin\theta$. $s^1 \in \mathbb{R}^2: x^2+y^2=1$

$$\therefore R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$\therefore$ the group $SO(2)$ is smoothly identified with s^1

$\hookrightarrow$ on single theta parameters.

2.) $\boxed{R \hat{\omega} R^T = \hat{\omega}}$ To prove;

Considering

$$\hat{\omega} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \text{ in a way that } \hat{\omega} \in SO_2$$

as $\hat{\omega}^T = -\hat{\omega}$ {Skew symmetric matrix}

$\therefore$ Given on left

$$R \hat{\omega} \Rightarrow \text{for } 2 \times 2 \text{ matrix of } R$$

we get

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\therefore R \hat{\omega} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} = \begin{bmatrix} -\omega \sin\theta & -\omega \cos\theta \\ \omega \cos\theta & -\omega \sin\theta \end{bmatrix}$$

$$R \hat{\omega} R^T = \begin{bmatrix} -\omega \sin\theta & -\omega \cos\theta \\ \omega \cos\theta & -\omega \sin\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 0 & -\omega(1) \\ \omega(1) & 0 \end{bmatrix} = \hat{\omega} = \text{RHS}$$